



Accelerating Agentic and Generative AI with SambaNova

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Director Public Sector Accounts

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- 1 Background
 - 2 Architecture Overview
 - 3 On-Premises and Cloud Deployments
 - 4 AI Use Cases

Who We Are

Snapshot

- Founded in 2017 by industry luminaries and originated at Stanford University
- Fully integrated generative AI platform, from 4th generation hardware to pre-trained models
- \$1B+ funding raised
- We are over approx. 400 employees with offices in Palo-Alto (HQ), Austin, London, Bangalore, Tokyo, Singapore, Australia

Founded by pioneers in AI



Clyde Hosein
President & COO



Rodrigo Liang
Co-founder & CEO



Kunle Olukotun
Co-founder &
Chief Technologist
& Stanford Professor



Christopher Ré
Co-founder &
Stanford Professor



Lip-Bu Tan
Chairman

Sophisticated, long-term investors

BlackRock
Capital Investment Corporation™

SoftBank
Partners Fund

TEMASEK

G/

intel
Capital

SAMSUNG
CATALYST
FUND

GIC

Micron

SK telecom

WALDEN
INTERNATIONAL

Celesta
Capital

LLM Model Trends

Models are getting bigger

- Llama 2 70B (2023)
- Llama 3 405B (2024)
- DeepSeek R1 671B (2025)
- Llama 4 Behemoth?
- DeepSeek R2?

Context Lengths are getting bigger

- Models were 4K just a few years ago
- Then 128K became the standard
- Now context lengths are growing to over 1M

Models are getting sparser

- Dense models were the norm (Llama 2 & 3 / GPT3)
- Then coarse Mixture of Experts (Mixtral / GPT4 both around 16 experts)
- Then fine grained Mixture of Experts (DeepSeek / Llama 4 both around 128+ experts)

Reasoning is getting unlocked by Test Time Compute

- All the best models today are reasoning models (DeepSeek R1, OpenAI o3)
- These models generate many more tokens, which means token generation speed is more important than ever

Market: Journey to Agentic AI Systems

Agentic AI increase inference calls per query by 100x

Key



Unique Model



Unique Step

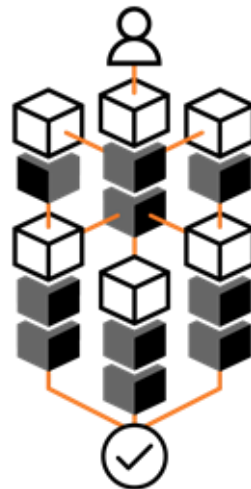
Simple Chatbot



Reasoning Models



Agentic System



- Agentic AI can increase inference calls per query by 100x

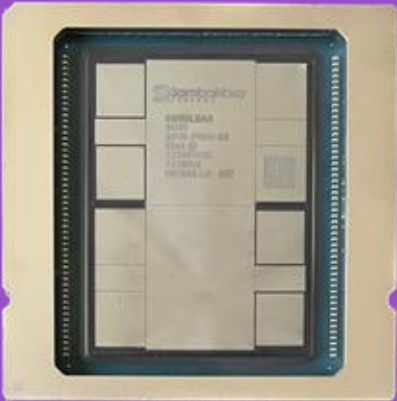
Tokens generated per query	1X	10X	Up to 100X
Models Used per query	1	1	~5-10
GPUs consume too many resources			

SN40L: SambaNova's 4th Gen AI Chip

Reconfigurable Dataflow Unit (RDU)

Native multi-tenancy support with fast model switching

Ideal for production inference, multi-tenancy, agentic workflows



sambanova
SN40L RDU

3-tier Dataflow Memory

520 MB On-Chip
SRAM Memory

Very fast memory for high speed inference with caching

64 GB High
Bandwidth Memory

Switch between models in as little as 2 milliseconds

1.5 TB High Capacity
DDR Memory

Hold large number of models in memory

SN40L: Tile Architecture

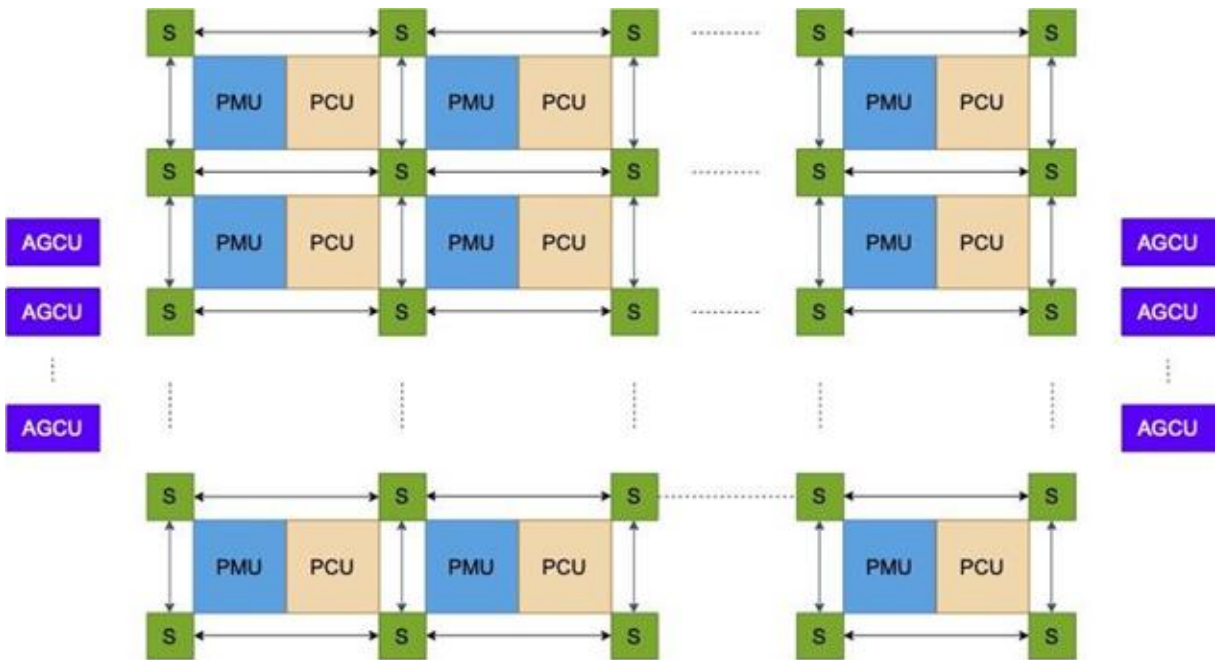
1040 PCUs and PMUs

PCU: Compute unit

PMU: Memory unit

S: Mesh switches

AGCU: Portal to off-chip memory and IO



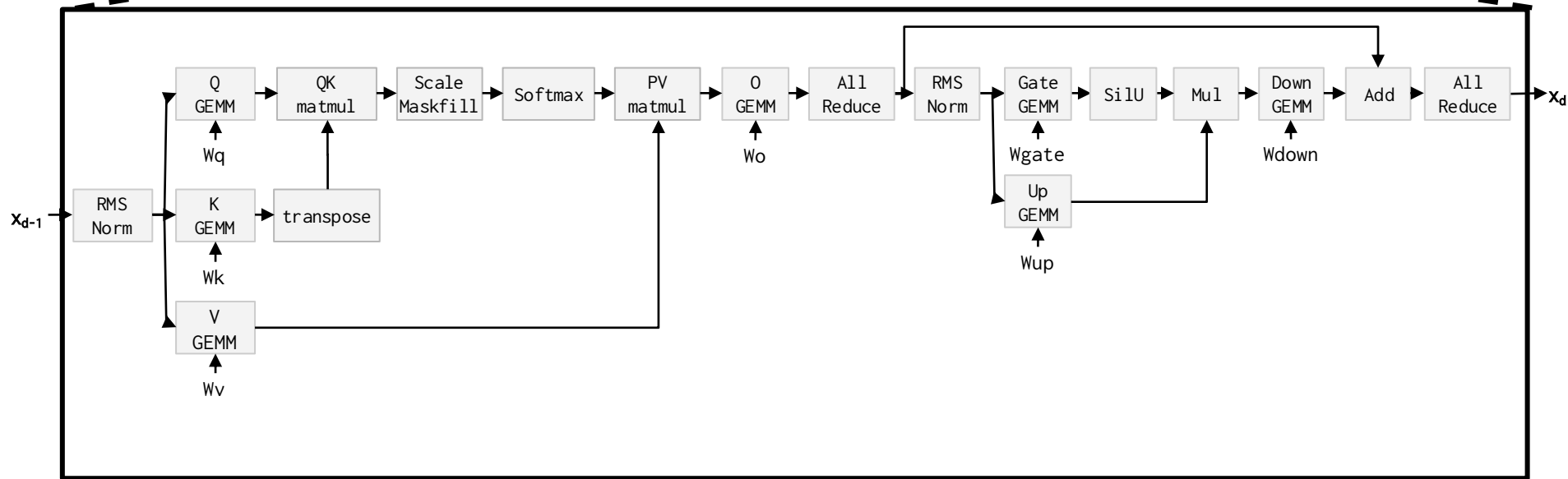
Structure of a Transformer

Example: Llama3.1 8B



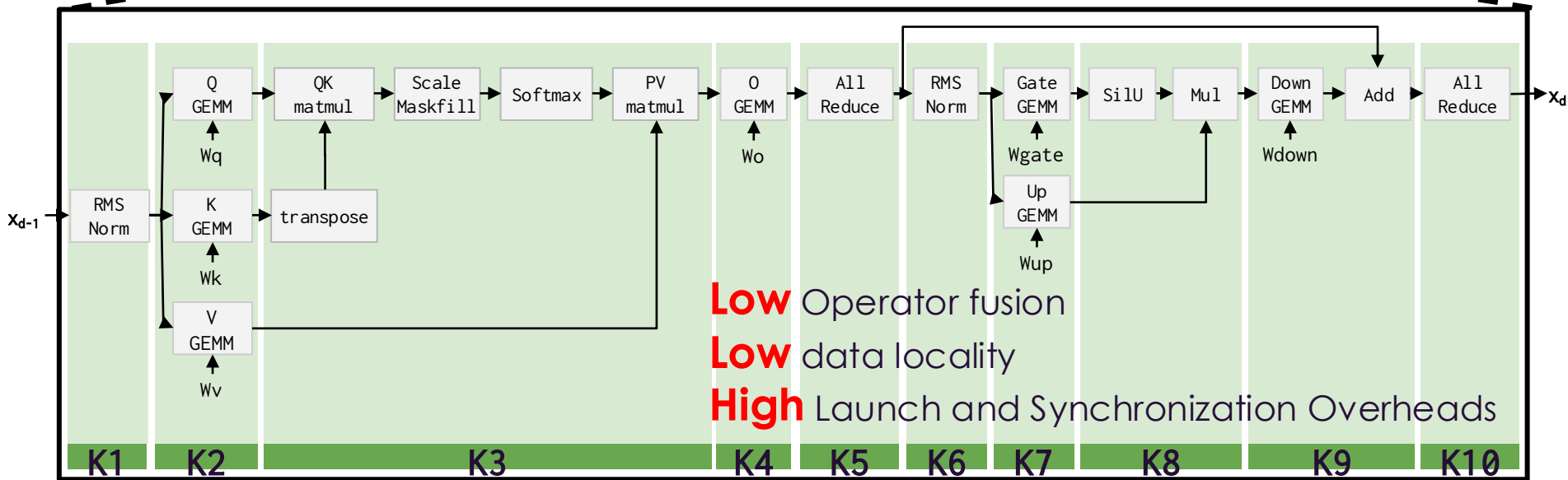
Structure of a Transformer

Example: Llama3.1 8B



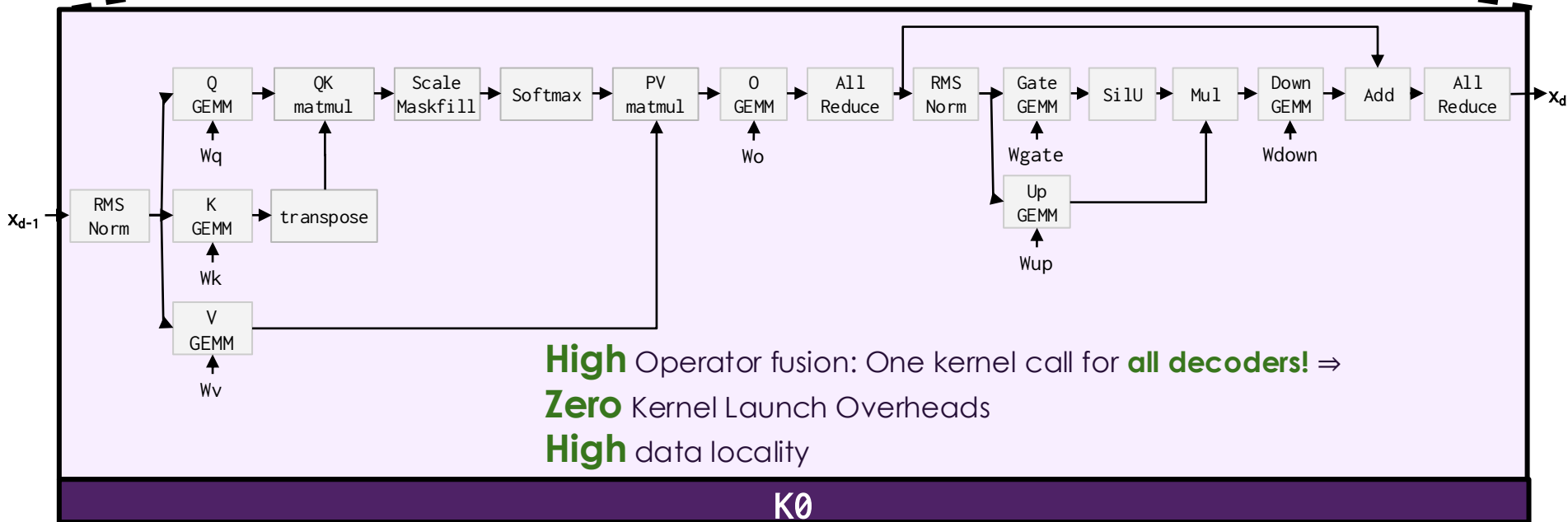
GPUs Incur Overhead with Low Operator Fusion

Example: Llama3.1 8B



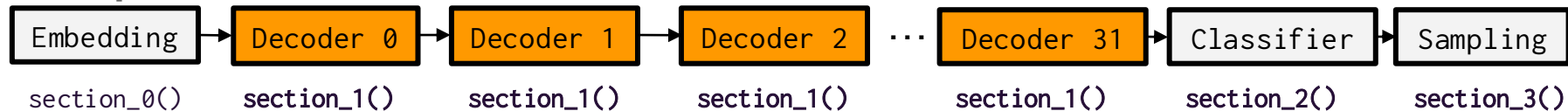
SN40L RDU Fuses the Entire Decoder into One Kernel!

Example: Llama3.1 8B



RDU Hardware Graph Orchestration: Temporal Fusion

Example: Llama3.1 8B



Single Section Decoder
+
Repeated Section Calls
=
New fusion opportunities!

```
section_1(arg):
    Foo;
    Bar;

// Repeat 32 times
section_1(arg0)
section_1(arg1)
...
section_1(arg31)
```

Kernel Looping

```
section_1(args[]):
    for i in range(32):
        Foo;
        Bar;

// 1 time
section_1(args)

1 kernel launch for all decoders!
```

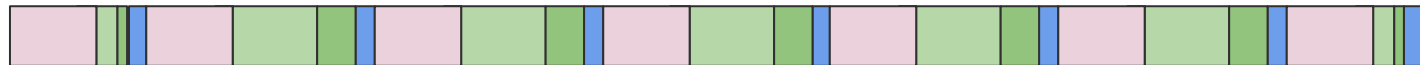
Executing Transformer on RDU

Example: Llama3.1 8B

Baseline: **100** tokens/s



+ Single Section Decoder

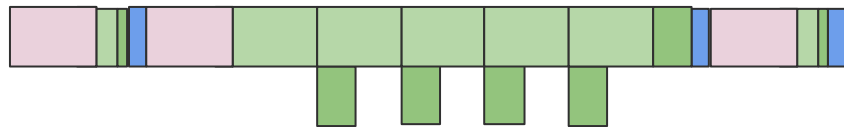


=

500 tokens/s
16 SN40L



+ Kernel Looping



=

1170 tokens/s
16 SN40L

Full Stack AI Inference Platform

Products:

- SambaStack – On Perm – you manage
- SambaManaged – On Prem - we manage
- SambaCloud – Our Cloud



Model Bundle, Spec Decoding, BYOC



Fully Integrated Rack-Level System
SambaRack



SambaNova's AI Accelerator Chip
Reconfigurable Dataflow Unit (RDU)



Performance

Optimized for inference

Sustainability

Most efficient energy consumption

Efficiency

10x fewer racks/chips than the competition

Inference Optimized

Supports custom checkpoints

Scalability Optimized

The building block for AI clouds and supercomputers

Rack Optimized

Comprehensive rack-level solution

SambaStack

Inference on SambaStack has a variety of features to serve your various use cases:

Model Bundling:

- Combine multiple models in a single OpenAI API compatible inference endpoint
- Run all models simultaneously, indicating the target model in each request

Speculative Decoding Pairs:

- Combine a main model with a draft model in a single endpoint to increase inference speeds

Bring Your Own Checkpoint (BYOC):

- Import external model checkpoints and run inference on them
- Imported model checkpoints can be externally fine-tuned or base models from Hugging Face

Self-Managed Through Standard Open-Source Tools:

- Kubernetes v 1.30+, Helm v 3.8.0+, and others

Model Bundling



Speculative Decoding Pairs



Bring Your Own Checkpoint

BYOC

SambaNova

Faster Inference on the Best Open Source Models



	SambaNova	Nvidia*
Llama 3.1 8B	932	180
Llama 4 Maverick	695	165
OpenAI's gpt-oss 120B	743	321
DeepSeek V3-0324	282	101
DeepSeek R1-0528	171	95

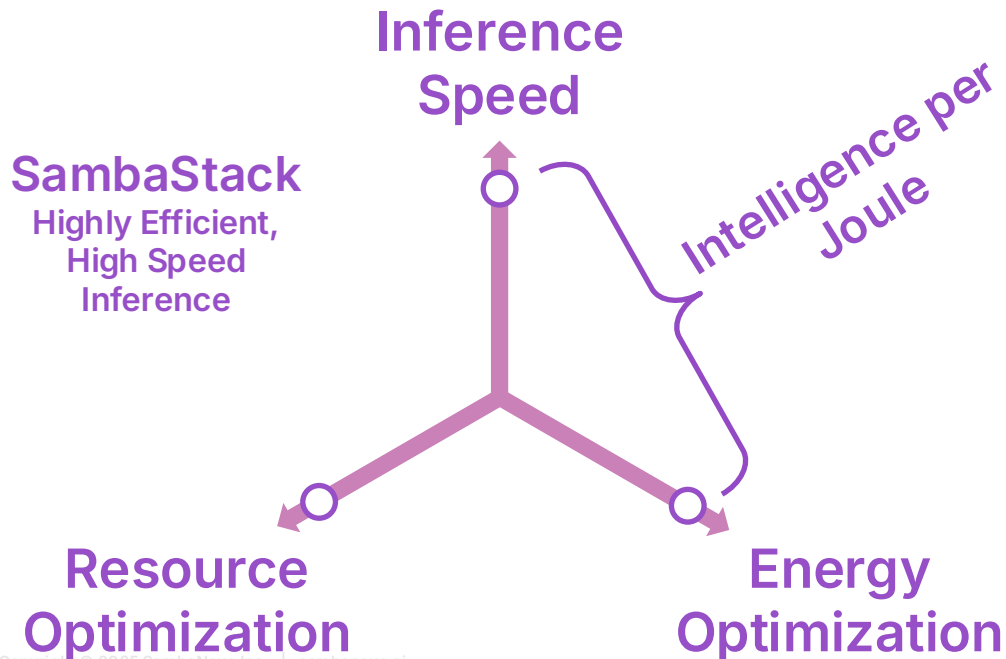
 Artificial Analysis

*Speeds benchmarked on Nvidia GPUs on Azure

Fast inference isn't just about speed -
it unlocks new types of applications and capabilities

SambaNova's SambaStack

Native multi-tenancy support with fast model switching
Ideal for production inference, multi-tenancy, agentic workflows



The Performance Measurement that Matters

Performance Metrics

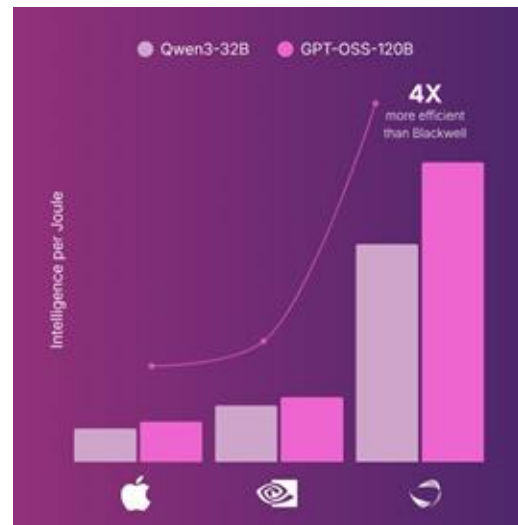
Tokens per second: Legacy metric that measures performance, but not quality of intelligence

Intelligence per joule: New metric that quantifies the end-to-end efficiency from a user's perspective, including the time cost of generation.

Provides a holistic view of the output of the entire AI system.

Measures the intelligence of the model, the efficiency of the software stack, and the energy of the hardware

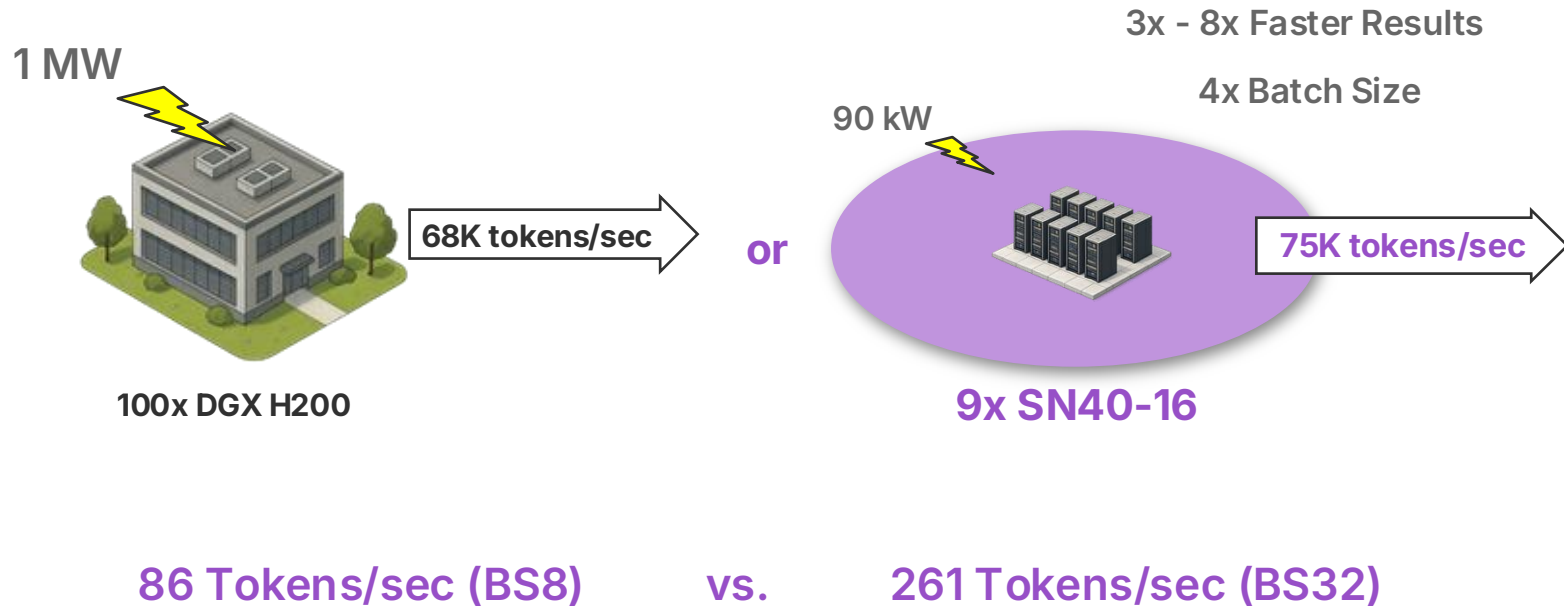
SambaNova SN40L is **4X** more efficient than B200



<https://arxiv.org/abs/2511.0788>

The Efficient AI Factory

Inference Performance (e.g. Llama 3.3-70B)



The 1 Megawatt Agentic Science Factory

Inference Performance (e.g. Llama 3.3-70B)

1 MW

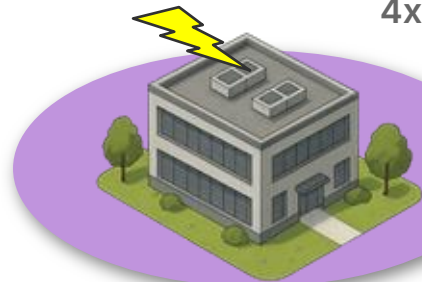


100x DGX H200

68K tokens/sec

or

1 MW



100x SN40-16

836K tokens/sec

3x - 8x Faster Results

4x Batch Size

86 Tokens/sec (BS8)

vs.

261 Tokens/sec (BS32)

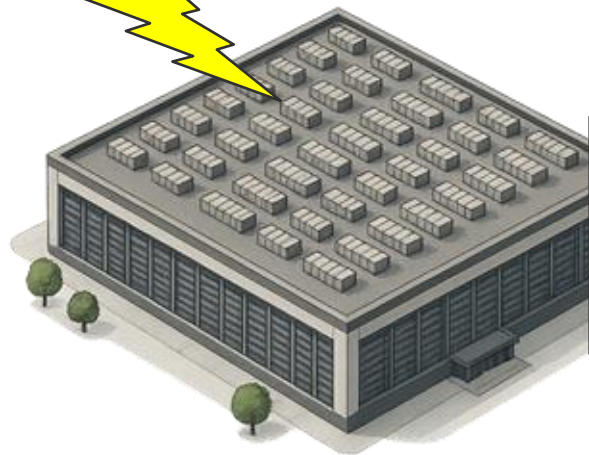
The Mega Token AI Factory (1 Million tokens/sec)

Mixed Workload - Inference Performance



Build New Infrastructure,
Significantly Expense,
Years to Reality

14.5MW



1M
tokens/sec



Existing Infrastructure,
Reality now

1.2MW



1M
tokens/sec

119x SN40-16

1453x DGX H200

A Better Architecture for Agents

Avg 10 kW

Model Hot-Swapping

Fast
Tokens/Second/User



Example Agents

Java Agent

Python Agent

SW Testing
Agent

Cloud Ops
Agent

Knowledge
Retrieval Agent

Contract Writing
Agent

Travel Booking
Agent

Dev Frameworks



LangChain



SambaStack Endpoint

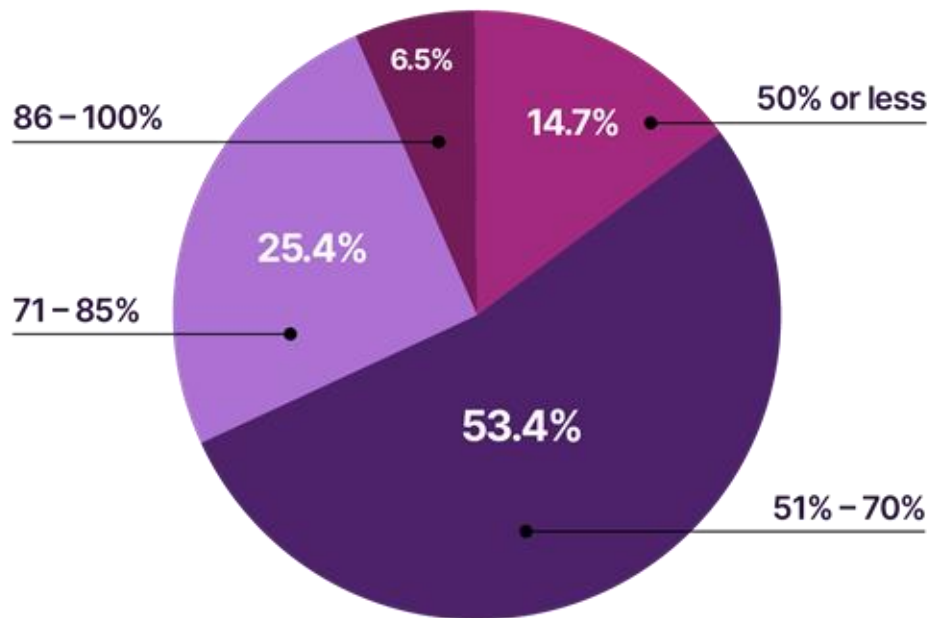
gpt-oss
120b

DeepSeek
V3-0324

Whisper
v3-Large

Meta
Llama 3.3
70B

More than 70% of organizations have underutilized GPUs



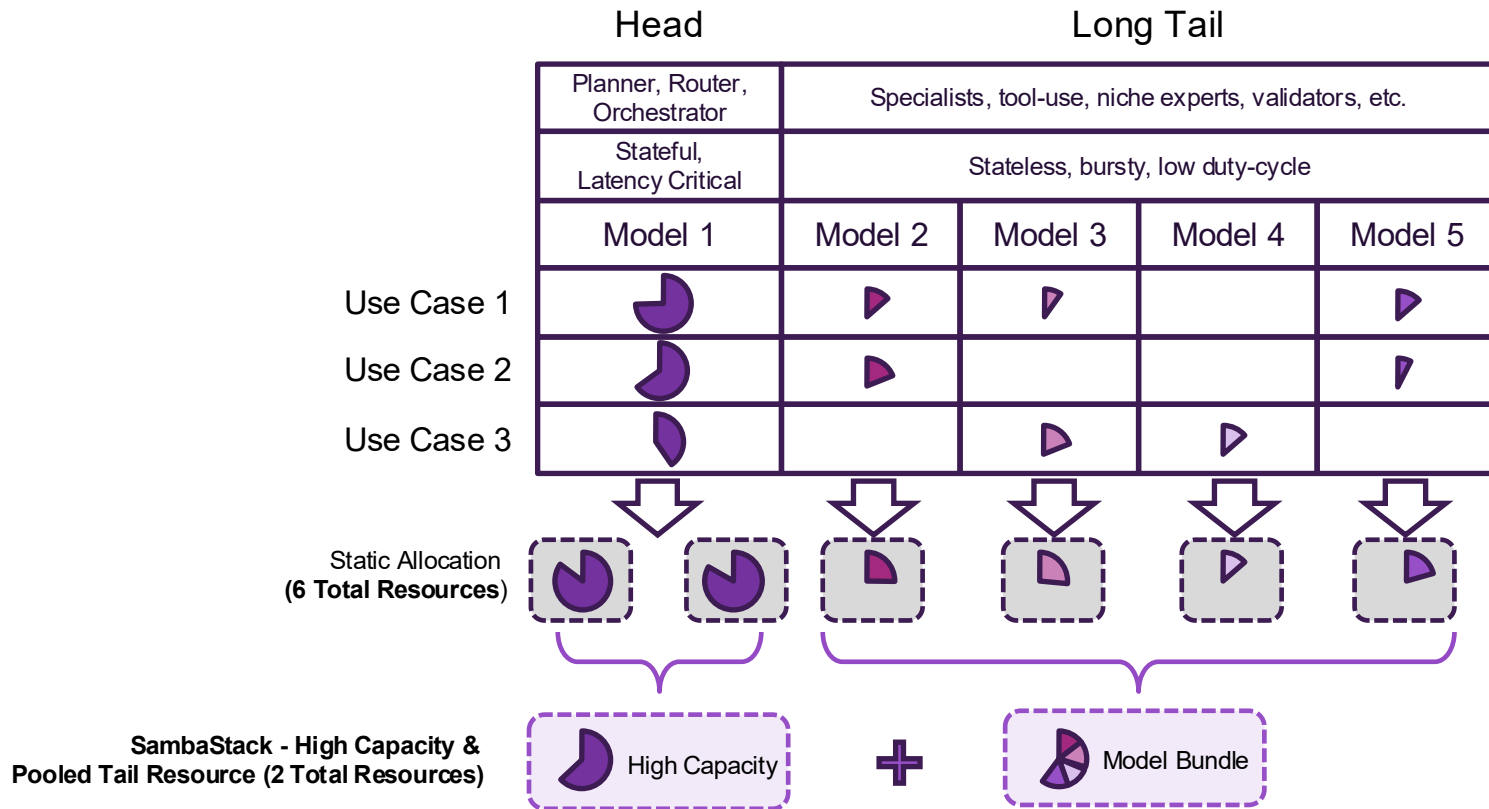
State of AI Infrastructure at Scale Survey

15% of respondents report that fewer than 50% of available GPUs are in use

53% of respondents believe less than 70% of their GPUs are utilized.

A better solution is required

Model Bundles – Agentic Workloads



SambaCloud

SambaCloud

Open source models available today



Llama 4-Maverick
Llama-3.3-70B-Instruct
Llama-3.3-Swallow-70B-Instruct-v0.4
Llama-3.1-8B-Instruct



Qwen3

Qwen3-235B
Qwen3-32B



deepseek

DeepSeek-R1 0528
DeepSeek-V3.1
DeepSeek-V3.2
DeepSeek-V3.1-Terminus
DeepSeek V3-0324
DeepSeek-R1-Distill-Llama-70B



OpenAI

GPT-OSS-120B
Whisper large v3



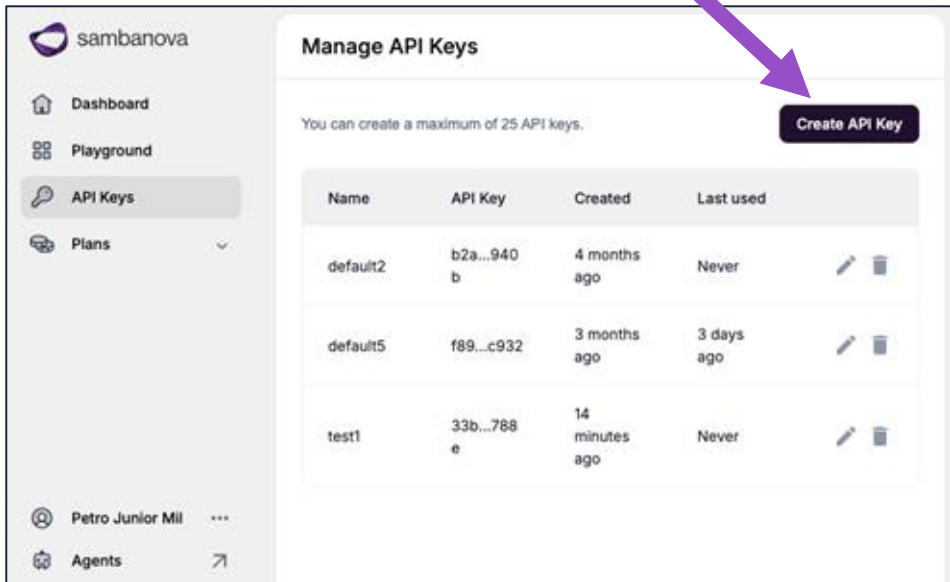
Mistral E5 7B

HUMAIN

ALLaM-7B

SambaNova is delivering the largest and most capable open source models, with unparalleled performance to unlock new capabilities that have been impossible to achieve

Get Your API Key at cloud.sambanova.ai

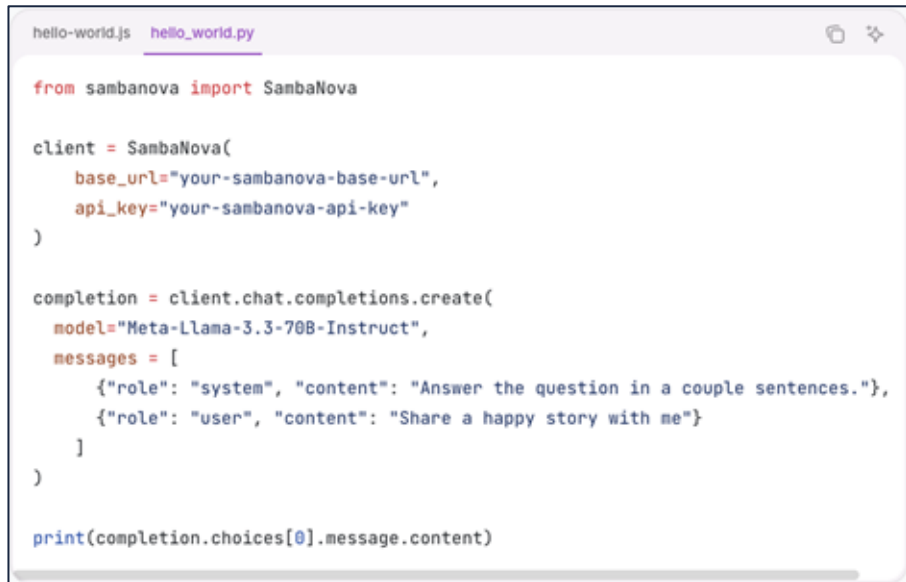


Manage API Keys

You can create a maximum of 25 API keys.

[Create API Key](#)

Name	API Key	Created	Last used	
default2	b2a...940b	4 months ago	Never	 
default5	f89...c932	3 months ago	3 days ago	 
test1	33b...788e	14 minutes ago	Never	 



```
hello-world.js  hello_world.py

from sambanova import SanbaNova

client = SanbaNova(
    base_url="your-sambanova-base-url",
    api_key="your-sambanova-api-key"
)

completion = client.chat.completions.create(
    model="Meta-Llama-3.3-70B-Instruct",
    messages = [
        {"role": "system", "content": "Answer the question in a couple sentences."},
        {"role": "user", "content": "Share a happy story with me"}
    ]
)

print(completion.choices[0].message.content)
```

 sambanova

[Dashboard](#)
[Playground](#)
[API Keys](#)

Plans

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[AI Starter Kits](#)
[Community](#)
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AI Starter Kits

[Developer Community](#)
[View all starter kits on GitHub](#)

Starter Kits Help You Build Fast – They bootstrap application development for common AI use cases with open-source Python code on SambaNova GitHub. They let you see how the code works, and customize it to your needs, so you can prove the business value of AI.

QuickStart - Cloud ReadMe

How to get started with SambaNova Cloud – A comprehensive guide to help you begin your journey with SambaNova Cloud.

[Getting Started](#)

Multi Modal Retriever

Chart, Image, and Figure Understanding – Unlock insights from complex PDFs and images with advanced retrieval and answer generation that combines both visual and textual data.

[Advanced AI Capabilities](#)
[Demo](#)

SambaAI Workspaces Integration

Seamless LLM Integration in Google Workspace – Enhance your productivity by integrating powerful language models directly into Google Docs and Sheets via App Scripts.

[Advanced AI Capabilities](#)

Llama 3.1 Instruct-o1

Enhanced Reasoning with Llama 3.1 405B – Experience advanced thinking capabilities with Llama 3.1 Instruct-o1, hosted on Hugging Face Spaces.

[Advanced AI Capabilities](#)
[Demo](#)

Function Calling

Tools calling implementation and generic function calling module – Enhance your AI applications with powerful function calling capabilities.

[Advanced AI Capabilities](#)
[Demo](#)

Enterprise Knowledge Retrieval

Document Q&A on PDF, TXT, DOC, and more – Bootstrap your document Q&A application with this sample implementation of a Retrieval Augmented Generation semantic search workflow using the SambaNova platform, built with Python and a Streamlit UI.

[Intelligent Information Retrieval](#)
[Demo](#)

Search Assistant

Include web search results in responses – Expand your application's knowledge with this implementation of the semantic search workflow and prompt construction strategies, with configurable integrations with multiple SERP APIs.

[Intelligent Information Retrieval](#)
[Demo](#)

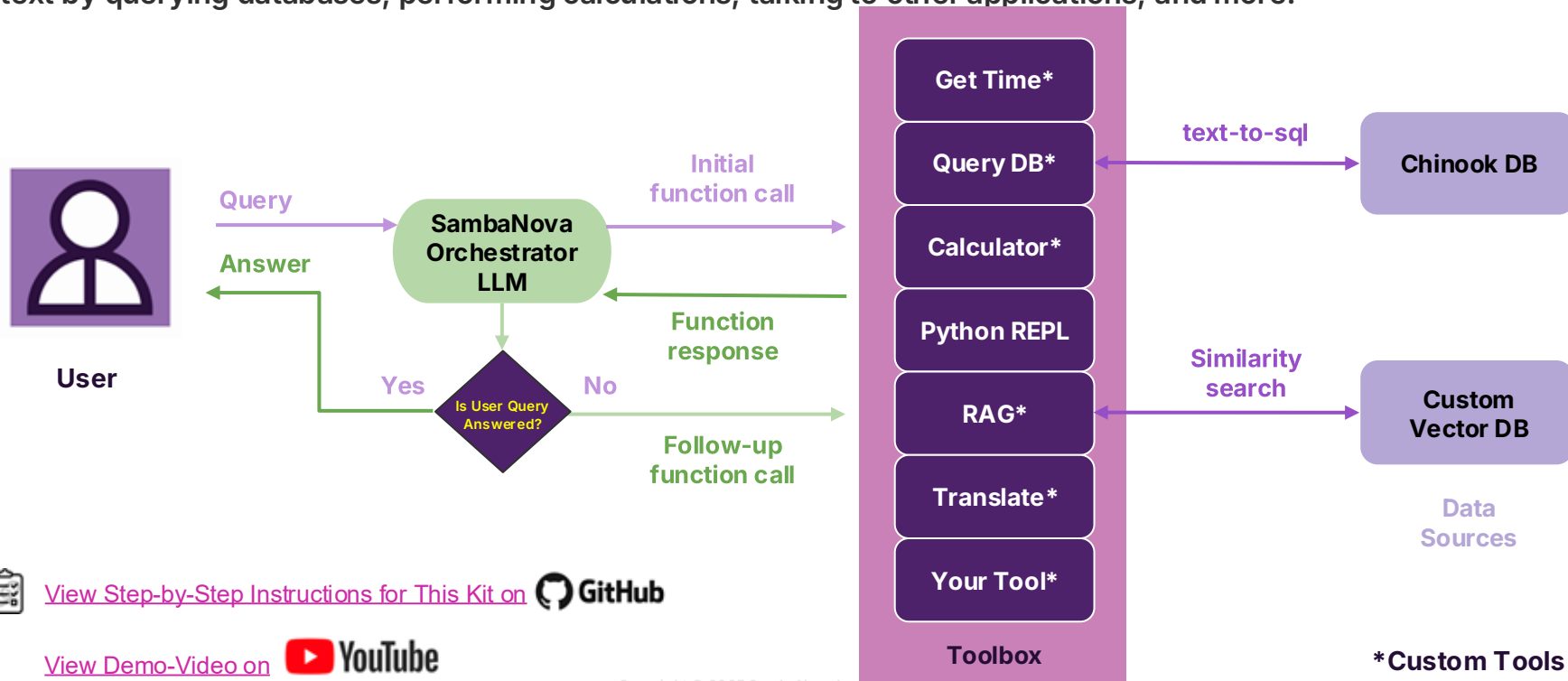
Benchmarking

Compare model performance – Quickly determine which models meet your speed and quality needs by comparing model outputs, Time to First Token, End-to-End Latency, Throughput, Latency, and more with configuration options in a chat interface.

[Model Development & Optimization](#)
[Demo](#)

Tool/Function Calling

Easily enable business use cases by adding function calls into NLP applications that do more than understand text by querying databases, performing calculations, talking to other applications, and more.



[View Step-by-Step Instructions for This Kit on](#)  **GitHub**

[View Demo-Video on](#)  **YouTube**

Agentic Applications



Input Objective

User provides high-level goals



Planning

Agent decomposes goals into tasks
1 Large LLM



Execution

Agent performs or delegates tasks
Many Small LLMs



Feedback and Learning

System improves through iteration

✓ Reduces manual intervention

✓ Increases efficiency and speed

✓ Enables complex multi-step task automation

[Agents Demo \(Try it yourself!\)](#)

Integration with External Frameworks / Platforms

LLM Frameworks



Hugging Face



LangChain



LiteLLM



LlamaIndex

Agent Frameworks



LangGraph



Semantic Kernel

Coding Assistants



CURSOR



Low-code platforms, vector DBs, and voice libraries



Langflow



milvus



hume



For complete list, visit our [Integrations](#) page

THANK YOU

tim.clarke@sambanova.ai

Try It for free!

<https://cloud.sambanova.ai>



Papers about SN40L

<https://arxiv.org/abs/2405.07518>



*"SambaNova SN40L:
Scaling the AI Memory
Wall with Dataflow and
Composition of Experts",
MICRO '24*

<https://arxiv.org/abs/2410.23668>



*"Kernel Looping:
Eliminating Synchronization
Boundaries for Peak
Inference Performance"*